OPEN AGENT HACKATHON 2026 · GENAI WORKS · TRACK 02 — THE AGENT THAT CAN EXPLAIN WHY

WhyLedger

Ask why a business number moved. Get the answer, what was ruled out, and a **receipt for every claim**.

Data · Zetaris Lightning | Memory · Meterless H-MEM + World Model | Model · NVIDIA Nemotron 3.5 Lightning | Built with Cursor agents

Team Charlie Agents (solo) · github.com/777sun77/whyledger · Apache-2.0

PROBLEM & TARGET USERS

“Why did the number move?” takes half a day — and the answer can't show its working.

The pain

- Explaining a KPI change means manually querying many **disconnected systems**: sales, returns, customers, promotions.
- The final slide rarely shows **its evidence** — which query, which rows.
- It almost never shows **what was ruled out**, so the same suspects get re-checked every time.
- LLM “explanations” make it worse: fluent text with numbers nobody can trace.

Who it's for

- **Retail & store operations managers** who must explain revenue swings.
- **FP&A / business analysts** who must explain KPI changes to leadership.

Example questions: *“Why did store revenue change from 2001 Q4 to 2002 Q4?”* · *“Why did yellow taxi trips drop from 2024-10 to 2025-02?”*

THE SOLUTION

An investigator agent that answers like an auditor would.

ANSWER

What explains it

Every hypothesis is tested with one read-only SQL query on Zetaris and scored **supported / partial / ruled out**, with the share of the movement it accounts for.

RULED OUT

What it wasn't

Rejected hypotheses get their own panel, with receipts — so *"we checked returns: not it"* is visible, not lost.

RECEIPTS

Proof for every claim

Each claim cites a receipt id (e.g. `ev-1`). Click it to see the exact SQL sent to Zetaris, the rows returned, timing and data lineage.

Numbers never come from the model. NVIDIA Nemotron frames, forms a hunch and narrates — but it only receives already-computed findings, and its answer must cite receipt ids that exist, or a deterministic template is shown instead.

It learns. Past investigations are recalled from Meterless H-MEM; 👍/👎 feedback goes into H-MEM's append-only trust ledger and shapes the next recall.

A REAL INVESTIGATION ON ZETARIS (TPC-DS RETAIL SAMPLE)

“Why did store revenue change from 2001 Q4 to 2002 Q4?”

STORE REVENUE

$402.33M → $407.50M

+$5.17M (+1.3%)

Ruled out — tested, not the cause

- × Demand moved to another channel
- × Shoppers from a few states
- × Promotions changed
- × Returns ate into it

Each line above and right is backed by a receipt: the exact SQL and rows from Zetaris (9 receipts in this run).

What explains it

SUPPORTED **Mostly volume.**
Volume +1.0%, value per unit +0.2% → volume effect +$4.20M, price effect +$964.3k · accounts for 81% of the movement

SUPPORTED **A customer segment shifted.**
College +$2.54M, Advanced Degree +$1.77M = 65% of gains

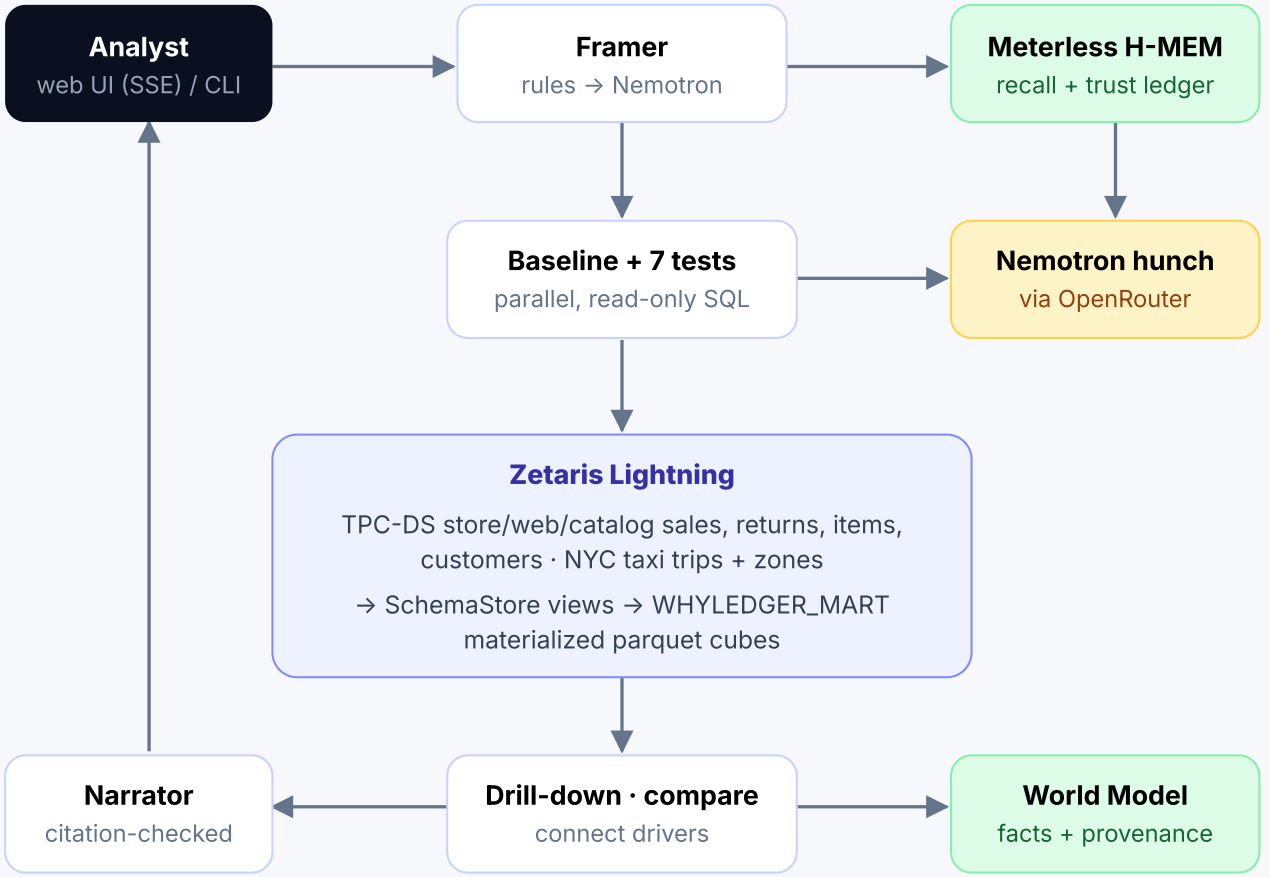
CONNECTED **College × category drill-down.**
Sports +$533.1k, Women +$517.7k = 40% of College gains

SUPPORTED **Product categories.**
Children +$1.79M, Electronics +$1.22M = 53% of gains

AGENT WORKFLOW & ARCHITECTURE

Frame → recall → measure → test in parallel → connect → narrate → learn

1. **Frame** the question: dataset, metric, periods, filters (rules → Nemotron).
2. **Recall** earlier investigations & feedback — Meterless H-MEM.
3. **Measure** the change: one Zetaris query → receipt `ev-1`.
4. **Test 7 hypotheses in parallel**: price vs volume, segment mix, geography, promotions, channel shift, returns, surcharges.
5. **Connect**: drill the strongest driver into the next dimension.
6. **Compare** Nemotron's up-front hunch with the evidence — receipts win.
7. **Write the evidence graph** to Meterless World Model, with provenance.
8. **Narrate** under a citation check, then **remember** the conclusion.



SPONSOR TECH · ZETARIS (REQUIRED)

Zetaris Lightning is the whole data layer — every number is a Zetaris query.

① Federate fragmented sources

`CREATE SCHEMASTORE VIEW`s union store, web & catalog sales and join dates, items, customer addresses, demographics (`01`), returns (`02`), and NYC trips × taxi zones (`06`). WhyLedger sees one semantic layer.

② Materialize investigation cubes

Lightning database `WHYLEDGER_MART` with `CREATE LIGHTNING FILESTORE TABLE … FORMAT PARQUET` (`03–07`). Hypothesis queries dropped from **15–45 s** (raw federated scans) to **under a second** each.

③ Receipts with lineage

Read-only SQL via the Lightning SQL API (`src/data/zetaris.ts`); a guard rejects anything but `SELECT` / `WITH`. Exact SQL, rows, timing and lineage are shown in the evidence drawer.

```
Lineage WHYLEDGER_MART.sales_cube ← schemastore view whyledger.sales_cube ← TPCDS_DB.store_sales + web_sales + catalog_sales JOIN date_dim, item, customer_address, customer_demographics
```

SPONSOR TECH · METERLESS (REQUIRED)

Meterless engines give the agent a memory and an auditable evidence graph.

H-MEM — explainable memory

- **MemoryRetrievalService**: recall with a retrieval trace before every investigation
- **MemoryStoreService**: each conclusion is stored (e.g. “stored as mem-7”)
- **TrustLedgerService**: 👍/👎 feedback into an append-only ledger (`/api/feedback`, `/api/memory/audit`) that shapes the next recall

World Model — provenance graph

- Event-sourced graph: question → *tested* → hypothesis → *driven_by* → segment (`upsertEntity`, `relate`)
- Every `assertFact` carries provenance: backend, SQL and a **sha-256 checksum** of the query and its rows
- Audit log shown in the UI

What's unique about our sponsor use

Zetaris isn't just a data source — it **is the evidence**: federation and parquet materialization make a 7-hypothesis investigation fast enough to run interactively, and every query becomes a receipt. Meterless isn't a chat memory — **H-MEM's trust ledger turns analyst feedback into memory confidence**, and **World Model provenance checksums** tie each fact back to the exact Zetaris query and rows.

PRODUCT VISUALS · RETAIL RUN ON ZETARIS

Answer, ruled-out panel, and the receipt behind every claim

STORE REVENUE · 2001 Q4 → 2002 Q4

$402.33M → $407.50M +$5.17M (1.3%) ev-1

Store revenue rose +$5.17M from 2001 Q4 to 2002 Q4. Volume growth (+1.0%) drove +$4.20M; price per unit (+0.2%) added +$964.3k, making volume the main factor [ev-6]. Customer segment shifts contributed significantly: College-educated shoppers +$2.54M and advanced-degree shoppers +$1.77M = 65% of gains, partly offset by -$1.45M elsewhere [ev-4]. Purchases in Children's and Electronics categories added $1.79M and $1.22M, comprising 53% of gains [ev-3]. Ruled out: channel shift (web moved opposite, only offsets 12%) [ev-7], state spread across 53 segments [ev-2], promoted-channel changes across 2 segments [ev-5], and returns movement [ev-8].

Next check: Identify which specific product categories or regions accounted for the +$4.20M volume gain.

narration: openrouter:nvidia/nemotron-3.5-lightning:free · 9 receipts · total 71.7s (SQL tests 6.4s)

👍 Helpful 👎 Not helpful Feedback updates the memory's trust ledger.

WHAT EXPLAINS IT

Price vs volume SUPPORTED

volume +1.0%, value per unit +0.2% — volume effect +$4.20M, price effect +$964.3k: mostly a volume story ev-6

accounts for 81% of the movement in that direction

A customer segment shifted SUPPORTED

College +$2.54M, Advanced Degree +$1.77M = 65% of all gains vs 28% of baseline revenue; partly offset by -$1.45M elsewhere ev-4

accounts for 65% of the movement in that direction

Connecting: College × category CONNECTED

College (+$2.54M) broken down by category: Sports +$533.1k, Women +$517.7k = 40% of all gains vs 19% of

Answer card + "What explains it" with supported / connected scores and receipt chips

Close ×

ev-6 · Price vs volume

ran on **Zetaris Lightning** in 6352 ms · 10/22/2026, 8:58:07 PM

Lineage · WHYLEDGER_MART.sales_cube ← schemastore view whyledger.sales_cube ← TPCDS_DB.store_sales + web_sales + catalog_sales JOIN date_dim, item, customer_address, customer_demographics

EXACT SQL SENT TO ZETARIS LIGHTNING

```
SELECT SUM(CASE WHEN (yr = 2001 AND qoy = 4) THEN revenue ELSE 0 END) AS v_from,
  SUM(CASE WHEN (yr = 2002 AND qoy = 4) THEN revenue ELSE 0 END) AS v_to,
  SUM(CASE WHEN (yr = 2001 AND qoy = 4) THEN units ELSE 0 END) AS u_from,
  SUM(CASE WHEN (yr = 2002 AND qoy = 4) THEN units ELSE 0 END) AS u_to
FROM WHYLEDGER_MART.sales_cube
WHERE ((yr = 2001 AND qoy = 4) OR (yr = 2002 AND qoy = 4)) AND channel = 'store'
```

ROWS RETURNED (1)

v_from	v_to	u_from	u_to
402,333,788.11	407,500,391.95	11,824,578	11,948,083

Receipt for "Price vs volume": lineage, exact SQL sent to Zetaris Lightning, rows returned

PRODUCT VISUALS · SECOND DATASET (NYC YELLOW TAXI)

Same agent, new domain: NYC taxi trips, with a live agent timeline

TRIPS · 2024-10 → 2025-02

3.83M → 3.58M -256.3k (-6.7%) ev-1

The 256.3k trip drop from 2024-10 to 2025-02 is driven almost entirely by the [time_band] factor: midday [-120.8k] and evening peak [-53.1k] account for 68% of the decline [ev-6]. All other dimensions were examined and ruled out as primary causes—[borough], [airport], [distance_band], and [payment] each show movement spread roughly in proportion across their segments, so they are not the source [ev-5 , ev-3 , ev-4 , ev-2]. The decline is independent of borough boundaries [ev-7].

Next check: Look at weekday vs weekend patterns within the midday and evening bands to see if a specific day-of-week shift is driving the midday drop.

narration: openrouter:nvidia/nemotron-3.5-lightning:free · 7 receipts · total 72.2s (SQL tests 5.4s)

👍 Helpful 👎 Not helpful Feedback updates the memory's trust ledger.

WHAT EXPLAINS IT

A time of day drove it SUPPORTED

midday -120.8k, evening peak -53.1k = 68% of all declines vs 58% of baseline revenue ev-6

accounts for 68% of the movement in that direction

Trips 3.83M → 3.58M (−6.7%): time band drove it; boroughs, airport, distance, payment ruled out · Nemotron narration passed the citation check · 72.2 s total

AGENT TIMELINE

13.7s	test	time_band: supported — midday -120.8k, evening peak -53.1k = 68% of all declines vs 58% of baseline revenue [ev-6 , 5367 ms]
24.5s	connect	midday (-120.8k) broken down by borough: spread evenly — the two drivers are independent [ev-7]
43.2s	hunch	Nemotron guessed "time_band" first — evidence agrees
43.5s	world-model	6 findings + 7 receipts written with provenance
72.2s	answer	openrouter:nvidia/nemotron-3.5-lightning:free
72.2s	remember	conclusion stored in H-MEM as mem-6

Agent timeline: test → connect → Nemotron hunch (“time_band first — evidence agrees”) → World Model write → answer → remember in H-MEM

EVALUATION · DOES IT EXPLAIN CORRECTLY?

We hid causes in real data. It found 16 of 17, and blamed nothing when nothing was there.

Planted causes · npm run eval:planted

A real TPC-DS quarter is copied as the next year's quarter **with a known cause planted**. Then the unchanged agent answers *"Why did store revenue change?"*

- 10/10 category, shopper-state and customer-segment causes: right hypothesis **and** right segment, no false blame
- 4/4 price rise · volume drop · store→web shift · returns spike
- 2/2 controls (noise only, uniform +6%): **nothing** blamed
- 1 miss promotion lift found, but a confounded "unknown" segment was also flagged. The drill-down shows the link.

16/17 scenarios · 75/76 checks

Faithfulness on real Zetaris · npm run eval:zetaris

All 5 example questions, live SQL and live Nemotron:

- **33/33** cited receipts exist
- **19/19** numbers in Nemotron's answers match the SQL results (new **numeric grounding guard**)
- Baselines from our semantic layer = **independent raw-table recompute, exactly** (402,333,788.11 → 407,500,391.95; 3,833,780 → 3,577,514 trips)
- 2 of 5 narrations were rejected by the guards (timeout, no citations), and the user got the template answer with the same receipts
- Full SQL investigation on Zetaris: **1.8 s** retail, **0.7 s** taxi

TECHNICAL STACK · NVIDIA · CURSOR

Small, typed, reproducible — and honest about the model's role.

Stack

- TypeScript · Node.js 20 · tsx
- Single-file web UI over server-sent events + CLI
- Zetaris Lightning SQL API · 7 SQL setup scripts
- Meterless H-MEM + World Model
- DuckDB TPC-DS sf0.1 offline mode (dev only)
- Unit tests (scoring, SQL guard, grounding) + `tsc` + 2 eval suites

NVIDIA Nemotron

- `nemotron-3.5-lightning` via the **OpenRouter free tier** (NVIDIA NIM provider also wired)
- Frames unusual questions, forms a hunch in parallel with SQL, narrates under citation + numeric grounding checks
- Disk-cached and time-budgeted — a slow model never blocks an answer

Built with Cursor agents

- Cursor-based AI coding agents wrote most of the code and ran it against a real Zetaris instance
- Real-data fixes found by agents: response shape, single-state stores, triple-counted taxi zones, slow scans → parquet cubes
- One human set direction, reviewed and approved

Why answers can be trusted: numbers come only from SQL · attribution is bounded (shares ≤ 100%, changes <1% capped at “partial”) · offsetting moves labelled co-movement, not proof of switching · rejected hypotheses reported with receipts.

FUTURE ENHANCEMENTS

Next: more sources, deeper follow-ups, fully local.

Live external sources

Add REST sources with Zetaris `CREATE LIGHTNING REST TABLE` (e.g. weather or public statistics) for cross-source “why”.

Follow-up questions

“Drill into Sports in Q4” — conversations that compound on H-MEM memory of earlier conclusions.

Guarded LLM-proposed drill-downs

Let Nemotron propose new hypothesis SQL, still behind the read-only guard and receipts.

Fully local models

Run narration on Meterless Runtime / Nightshift's OpenAI-compatible endpoint; World Model graph panel in the UI.

WhyLedger

Receipts for every claim.

`github.com/777sun77/whyledger`